

A Versatile Framework for Formula-Based Enforcement and Synthesis in Abstract Argumentation

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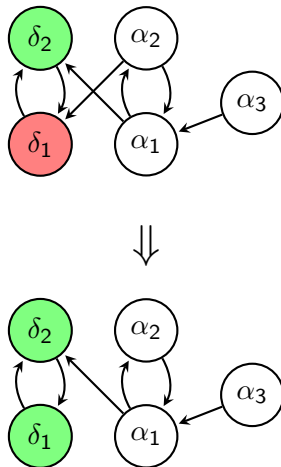
How does one revise an argumentation framework?

Motivation

- Argumentation-based AI systems may need to be revised when new evidence contradicts the current conclusions.
- Various existing algorithmic approaches to **enforcement** in **abstract argumentation**: focus on minimizing **syntactic change**.

Key problem

Minimizing changes to the structure of the framework does not necessarily minimize changes to **acceptability statuses** of arguments.



Can we minimize semantic and syntactic change simultaneously?

Formula-based enforcement

A formula-based enforcement instance specifies **propositional constraints** which should be satisfied by the resulting argumentation framework.

Φ^+ : desired behavior

Formulas that should hold in **some extension**.

$$in_a \wedge in_b$$

Φ^- : forbidden behavior

Formulas that should hold in **no extension**.

$$\neg in_b \vee \neg in_c$$

Φ^R : structural constraints

Formulas constraining the **attack relation**.

$$att_{a,b} \rightarrow att_{b,c}$$

Hard and soft constraints

Weights on formulas distinguish requirements that **must** be satisfied from **preferences**.

Constraints on semantic and syntactic change can be combined into a single instance.

Generality of formula-based enforcement

Formula-based enforcement captures various existing enforcement notions as special cases:

- extension enforcement (Baumann & Brewka, 2010; Coste-Marquis et al., 2015; Wallner et al., 2017)
- status enforcement (Niskanen et al., 2016)
- rejection enforcement (Heine & Ulbricht, 2024)
- argumentation framework synthesis (Niskanen et al., 2019)

New enforcement variants

For example:

$$in_a \xrightarrow{w} \infty, \quad in_b \xrightarrow{w} \infty, \quad in_a \wedge in_b \xrightarrow{w} k > |A|^2$$

Require a and b to be individually accepted, while **preferably accepting them together**.

Complexity of formula-based enforcement

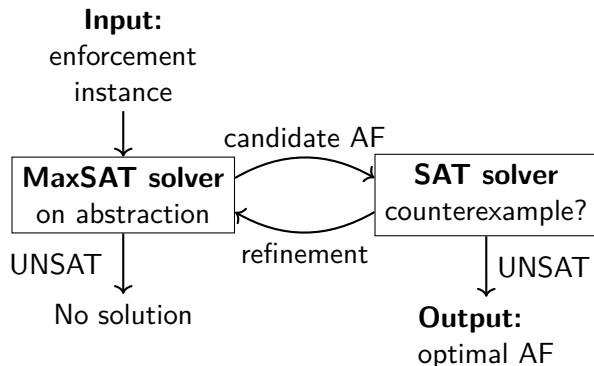
Expressivity of formula-based enforcement comes at a price: high computational complexity.

Semantics	Complexity
Grounded	NP-complete
Admissible	Σ_2^P -complete
Complete	Σ_2^P -complete
Stable	Σ_2^P -complete
Preferred	Σ_3^P -complete

Can we still solve practical instances exactly?

MaxSAT-based counterexample-guided abstraction refinement (CEGAR)

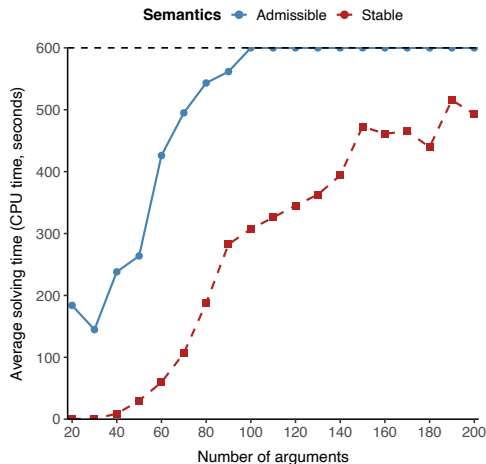
Exact algorithm for Σ_2^P -complete variants: instead of encoding the full problem at once, iteratively solve an easier abstraction and refine it when counterexamples are encountered.



High-level idea

- 1 Call a MaxSAT solver to find an optimal candidate attack structure.
- 2 Check using a SAT solver whether the candidate AF has a counterexample extension.
- 3 If a counterexample exists, refine the abstraction by adding a clause which excludes it.
- 4 Repeat until no counterexample remains.

How far can the algorithm scale?



Setup

- 950 generated instances.
- $n = 20, 40, \dots, 200$ arguments.
- Admissible vs. stable semantics.
- 600 s timeout per instance.

Observations

- Stable semantics scales to roughly 70 arguments.
- Admissible semantics requires far more CEGAR iterations:
 ≈ 2100 vs. ≈ 46 .

<https://bitbucket.org/andreasniskanen/phiaf>

Contributions

Formula-based enforcement: unified language for expressing semantic and syntactic constraints for enforcement in abstract argumentation via weighted propositional formulas.

- **Generalizes** various existing notions while enabling new forms of enforcement.
- **Complexity** ranges from NP-completeness to Σ_3^P -completeness.
- **Exact algorithm** for Σ_2^P -complete variants by employing MaxSAT-based CEGAR.
- **Empirical evaluation** demonstrates scalability to moderately-sized AFs.

Thank you for your attention! Questions?

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